Gaussian Splatting for Physical Inverse Problems — Meeting Brief

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Core idea: represent the unknown field as Gaussians, render them on a grid, and replace the camera of Gaussian Splatting with the physics operator; fit to the measurements. Data from InverseBench.
Code: `github.com/muhammadhussainqureshi/thesis-gs`

1. 3D Inverse Scattering

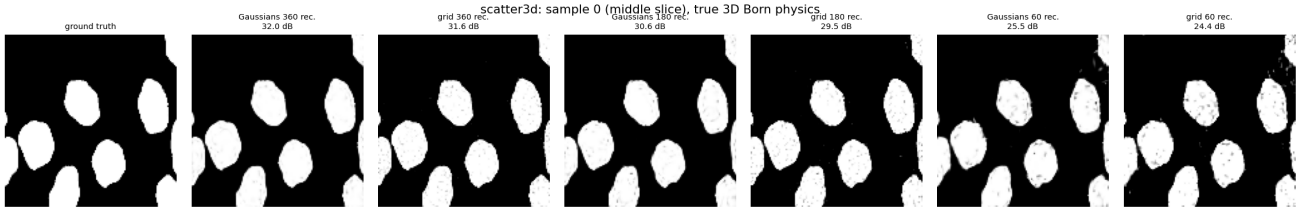
What I did, and how

- 3D Gaussians (position, scale, rotation, amplitude) reconstruct cell permittivity from scattered fields; compared with a voxel grid optimised with the same loss.
- **True 3D Born model** (your request): 3D Green's functions, 60 transmitters and 360 / 180 / 60 receivers on a sphere. Verified against the exact point-scatterer response (10^{-7}).
- Data: real 3D nuclei (BBBC024 / CytoPacq, $32 \times 128 \times 128$) and synthetic nuclei; also 2D InverseBench cells.
- Fixes found along the way: a sparsity term (removes "ghost" copies in the sensors' blind spots) and a **bounded density** $1 - e^{-f}$ (Gaussians could overshoot; the grid is clipped to $[0, 1]$).

Results (PSNR dB, true 3D model, same regularisation for both)

	Parameters	360 rec.	180 rec.	60 rec.
Real nuclei: Gaussians (bounded)	60k	**32.8**	**31.4**	**26.7**
Real nuclei: voxel grid	524k	32.5	30.5	25.6
Synthetic: Gaussians (bounded)	54k	**46.4**	**43.6**	**39.1**
Synthetic: voxel grid	524k	42.1	39.9	38.2

- Gaussians win on **every volume** at every receiver count, with **∼9× fewer parameters**.
- Without the bounded density they lost by 5–7 dB. With more measurements (slice-wise 3D, 2D) the grid is still slightly ahead (0.2–2.7 dB).



Questions

1. Is this true 3D (Born) model what you wanted? Is single scattering enough for Thesis I?
2. Can the claim be: *bounded Gaussians are more accurate and ∼9× more compact when measurements are scarce; a grid is better when data are plentiful*?

2. 4D Navier–Stokes

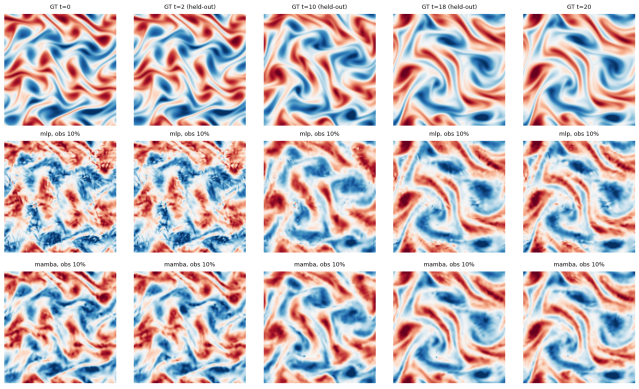
What I did, and how

- 4D Gaussians: canonical 2D Gaussians plus a **deformation network** that moves them over time (2D + time, 21 frames, $Re = 200$, InverseBench data).
- Deformation by an **MLP** (as in 4D Gaussian Splatting) vs. **Mamba** (state-space model over each Gaussian's trajectory) vs. static Gaussians; plus a temporal smoothness term.
- Observed 100% or only 10% of pixels; 5 frames never observed (tested).

Results (PSNR dB on unseen frames, 3 flows)

Observed	Static	MLP	Mamba
100%	17.0	**27.4**	27.2
10%	16.0	23.0	**25.4**

- Motion model: **+10 dB** over static.
- Sparse data: **Mamba +2.4 dB** over MLP, on all 3 flows; MLP shows streaks.
- Full data: equal. Mamba is ∼9× slower.



Rows: truth, MLP, Mamba (10% observed).

Questions

3. Is reconstructing the flow from sparse observations acceptable, or should I add physics (Navier–Stokes residual) or recover the initial condition through the solver, as InverseBench does?
4. Is Mamba as the motion model what you intended?

General questions

5. **Thesis I:** the exact list of targets, the required template, and the thesis title?
6. **Next 2 weeks:** physics-informed 4D model, more samples, or extending the literature review?